Shreya Abraham Varghese

sav.shreya02@gmail.com +44 7741 553151 Portfolio

Summary

MSc Advanced Computer Science (AI) graduate from the University of Manchester with hands-on experience in Python, data analysis, machine learning, and backend development. Experienced in building predictive models, data pipelines, REST APIs, and analytical applications using real-world datasets.

Profiles

Github
shreya-varghese

LinkedIn
Shreya Abraham Varghese

Experience

Krishna Kartik Homes September 2024 – Present

Junior Python Developer

- Developed and maintained the company's main responsive real-estate website using HTML, CSS, and JavaScript, including reusable property pages, enquiry forms, popup flows, and interactive tools.
- Built a backend lead-capture workflow using API endpoints and database storage to validate, store, and manage enquiries submitted through multiple website forms.
- Implemented a RAG-based document Q&A system using Python, embeddings, and semantic search, enabling retrieval across 1,000+ document chunks.
- Developed a lead-classification model to categorise enquiries as Hot, Warm, or Cold, helping prioritise sales follow-ups.
- Built internal tools including an EMI calculator and property-tax estimator to support real-estate sales conversations and client decision-making.

ServiceNow January 2026 – April 2026

Trainee – Rise Up with ServiceNow (AuraSync Project)

- Built an employee lifecycle management application automating leave delegation, internal transfers, and offboarding workflows.
- Developed custom tables, UI policies, user roles, and access-control rules within a scoped ServiceNow application.
- Implemented multi-step Flow Designer workflows for approvals, task reassignment, and lifecycle tracking and automated access provisioning and revocation processes to reduce manual effort and improve operational consistency.

CodeClause August 2023 – September 2023

Machine Learning Intern

Built a real-time face detection pipeline using Python, OpenCV, and Flask, achieving 25–30 FPS and improving accuracy by 20% through preprocessing and multi-scale detection. Deployed via REST API for low-latency inference.

Languages

English
Highly Proficient

Hindi
Native

Malayalam
Proficient

Projects

Intelligent Power Demand Forecasting System

Developed an end-to-end electricity demand forecasting system integrating historical consumption, weather data, and regional holidays. Performed EDA on 52,000+ observations and engineered 52 predictive features. Trained XGBoost models achieving test R^2 scores of 0.994–0.997 across three feeders. Built a FastAPI backend and interactive dashboard delivering 24-hour forecasts, containerised using Docker.

Carbon Emissions Analytics and Reporting Platform

Developed an end-to-end ESG reporting platform for Scope 1 and Scope 2 emissions. Designed a relational database with versioned emission factors, historical calculations, and audit logging. Implemented REST APIs for year-over-year comparisons, emission intensity, and hotspot analysis. Built an interactive dashboard with KPI visualisations and reproducible synthetic data for multi-year reporting.

Semi-Supervised Brain-Tumor Segmentation Using a Box-Constrained Loss in a 2D U-Net

Developed a lightweight MobileNetV2–U-Net pipeline for whole-tumor segmentation on MRI scans using a semi-supervised, mixed-annotation strategy. Designed a two-stage training process combining mask-supervised pre-training with bounding-box–guided fine-tuning. Introduced distance-map priors and outside-box penalties to learn from coarse labels, reducing reliance on costly voxel-level annotations. Delivered a reproducible pipeline with documented error analysis and scalability considerations.

Skills

Programming & Data Analysis: Python, SQL, JavaScript, Pandas, NumPy, Exploratory Data Analysis (EDA), Data Cleaning, Feature Engineering

Machine Learning & AI: Scikit-learn, XGBoost, Time-Series Forecasting, Predictive Modelling, Deep Learning, NLP, Computer Vision

Backend & APIs: FastAPI, Flask, REST APIs, SQLAlchemy, API Integration

Databases & Data Systems: SQLite, Database Design, Retrieval-Augmented Generation (RAG), Embeddings, Semantic Search

Visualisation & Reporting: Matplotlib, Seaborn, Interactive Dashboards, Data Visualisation

Tools & Deployment: Git, Docker, Jupyter Notebook, ServiceNow

Education

The University of Manchester — Masters of Science
Advanced computer science, Artificial Intelligence — September 2024 – December 2025

SRM Institute of Science and Technology — Bachelors in Technology
Computer Science — September 2020 – May 2024

Certifications

Fundamentals of Deep Learning
NVIDIA

Micro-Certification - Flows
ServiceNow

Machine Learning with Python
IBM

Data Analysis with Python
IBM

Python Project for Data science
IBM

Power BI Virtual Case Experience
PWC

AWS Academy Cloud Operations
AWS

Database Foundations
Oracle